Vertika Singh

Full Stack AI Developer & AI Engineer

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Professional Summary

AI-ML focused Software Developer with hands-on experience building production-grade AI agents, RAG pipelines, and full-stack applications.Proficient in **Python, FastAPI, Pydantic, LangChain, CrewAI, Next.js, PostgreSQL, and Docker**, with a strong foundation in software engineering and **Generative AI**.

Technical Skills

Languages & Frontend: JavaScript, TypeScript, Python, Java, React.js, Next.js, Tailwind CSS.
Backend & Databases: Node.js, Express.js, FastAPI, WebSockets, WebRTC, JWT, PostgreSQL, MongoDB, Redis.
AI/ML & LLMs: TensorFlow, PyTorch, scikit-learn, OpenCV, NLP, RAG, LangChain, LangGraph, Prompt Engineering, Ollama, LlamaIndex, OpenAI APIs, n8n Automation, CrewAI, Hugging Face, Pydantic
Cloud, DevOps & Tools: Git, GitHub, Docker, AWS (EC2, S3), Vercel, SDLC, Google Colab.

Experience

Schoollog- LB Microtechnologies Pvt. Ltd. **July 23, 2026 – October 23, 2026**
AI Engineer Intern *Remote*

- Built a multi-tenant voice-agent platform that answers phone calls in Hindi, English and Hinglish (**LiveKit, Next.js, Supabase** with row-level security, **Python** workers), running at about **$0.0065 per call-minute** for speech, AI and voice.
- Architected a multi-agent voice calling platform on **LiveKit**, enabling teams to spin up multiple calling agents, each grounded in its own knowledge base built from text, documents, and URLs.
- Halved the AI prompt from about 4,200 to about 2,000 tokens by searching the knowledge base per question, which cut reply time from **2.1 s to 1.8 s** in a simulated call.

Featured Projects

ORBIT- AI Freelance Workspace Live | GitHub
Personal Project – 2026

- **Tech Stack:** FastAPI, CrewAI, Celery, Redis, PostgreSQL, Supabase, Next.js, TypeScript.
- Built a production multi-agent system using CrewAI that automates the full freelance business workflow — brief parsing, scope pricing against Indian market rates, proposal drafting, GST-compliant invoice generation, and payment follow-up — reducing admin time from 10+ hours/week to under 30 minutes.
- Designed an event-driven automation layer using Celery + Redis where marking a milestone complete auto-generates an invoice and a daily cron drafts follow-up messages for overdue payments, eliminating all manual triggers
- Deployed a SaaS architecture with row-level security on PostgreSQL (Supabase), JWT auth, and Next.js — handling complete data isolation across freelancer accounts with zero cross-tenant leakage.

KOKO- Offline AI Agent GitHub
Personal Project – 2025

- **Tech Stack:** Python, Pydantic, Streamlit, Ollama, Llama 3.2, FastAPI, Moondream, SQLite.
- Engineered a local AI agent using Ollama (Llama 3B / Moondream 1.8B), ensuring zero data leakage for private system tasks.
- Built a custom agentic loop with deterministic routing and regex safety fallbacks to execute Python (psutil) and SQL commands accurately.
- Bypassed native JSON formatting limits of small models by engineering an XML-tagging pipeline to structured local SQLite data queries.

Mango- A Developer Workspace Live | GitHub
Personal Project – 2026

- **Tech Stack:** Next.js, React, TypeScript, Tailwind CSS v4, Supabase, PostgreSQL, MCP, Groq API, OAuth
- Built an AI-native developer workspace using Next.js, TypeScript, Supabase, and Groq, integrating Llama 3.3/Mixtral for context-aware project assistance.
- Implemented hybrid RAG with pgvector + keyword search to retrieve context from project documents, PRDs, notes, and tasks for grounded AI recommendations.
- Developed an MCP-based AI agent with structured tool execution, GitHub integration, and human-in-the-loop approvals for safe automated actions.

FaceScan- Offline Biometric Attendance Application GitHub
Personal Project – 2026

- **Tech Stack:** Dart, Flutter, Python, TensorFlow Lite, InsightFace Buffalo SC, MiniFASNet V1SE/V2, ML Kit Face Detection
- Engineered an offline biometric attendance system with Flutter + TensorFlow Lite, achieving 70 ms face-recognition inference using InsightFace Buffalo SC with 512-D embeddings.
- Developed a multi-layer anti-spoofing pipeline combining 2 MiniFASNet models, FFT-based screen detection, glare/bezel analysis, and temporal motion checks to detect photo and screen-based attacks.
- Implemented dual-factor GPS + Wi-Fi geofencing with AES-256 encrypted SQLCipher storage, enabling offline attendance verification while securing biometric embeddings and attendance records.

Education

JSS Academy of Technical Education **Expected Oct 2027**
B.Tech in Computer Science and Engineering (AI/ML) *Noida, Uttar Pradesh (201301)*